

Reinforcement Learning for Text Detoxification: Balancing Style Control and Semantic Preservation

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Abstract

Text detoxification aims to remove toxic language while preserving the original meaning. This study investigates whether reinforcement learning can improve detoxification beyond supervised fine-tuning without sacrificing semantic preservation. Starting from a LoRA-fine-tuned language model, two GRPO objectives are compared: a style-only reward and a combined style-semantic reward. Style-only optimization improves detoxification but causes substantial semantic degradation. Adding a semantic reward substantially improves semantic preservation over both SFT and style-only GRPO. However, it does not improve detoxification over SFT; its slightly lower observed detoxification success is not distinguishable from the SFT baseline in the paired test-set analysis. These results highlight the importance of reward design in balancing detoxification and semantic preservation.

1 Introduction

Text detoxification is a form of controlled text generation in which toxic, rude, or offensive language is rewritten into a neutral form while the underlying meaning is preserved. The task can be viewed as a form of text style transfer: toxicity should change, while semantic content should remain as stable as possible (Jin et al., 2022). Parallel resources such as ParaDetox (Logacheva et al., 2022) make supervised fine-tuning possible by providing toxic sentences paired with human-written neutral rewrites.

Supervised learning, however, optimizes token-level likelihood rather than the sequence-level properties used to evaluate a generated rewrite. Reinforcement learning offers an alternative by allowing generated sequences to be optimized directly according to task-specific reward functions. Recent work on language-model reasoning has demonstrated the effectiveness of Group Relative Policy Optimization (GRPO), which estimates relative

advantages from groups of sampled completions without requiring a separate value model (Shao et al., 2024). Although the algorithm was not introduced for detoxification, its group-based optimization framework can also be applied to controllable generation.

Direct reward optimization introduces an important risk. A model optimized only for low toxicity may discover outputs that satisfy the classifier while dropping, weakening, or changing the source meaning. Similar tensions between toxicity reduction and content preservation have been observed in previous work on detoxification (Welbl et al., 2021). This makes reward design particularly important: improving the target style alone does not necessarily imply successful rewriting.

This study therefore asks:

Can reinforcement learning improve detoxification in controllable language generation beyond supervised fine-tuning without sacrificing semantic preservation?

The question is investigated using Qwen2.5-0.5B-Instruct (Yang et al., 2024) on ParaDetox. A LoRA-based supervised model (Hu et al., 2022) serves as the main baseline. Two independent GRPO branches are then initialized from the same SFT checkpoint. The first optimizes only a toxicity-based style reward, while the second combines a normalized style reward with a sentence-embedding semantic reward.

The experiments reveal a clear reward-dependent trade-off. Style-only GRPO substantially improves detoxification but causes severe semantic degradation. Introducing semantic preservation into the reward restores and improves the automatic semantic-similarity metric while retaining a detoxification success rate close to the SFT baseline. The results therefore show that RL can strongly optimize detoxification, but that achieving this objective without semantic degradation depends critically on the reward formulation.

2 Related Work

Text detoxification and style transfer. Text style transfer modifies a stylistic attribute while attempting to preserve style-independent content (Jin et al., 2022). Toxicity removal is a particularly challenging instance because aggressive removal of offensive expressions can alter semantic content or erase information. ParaDetox provides a parallel corpus of toxic English sentences and human-written neutral paraphrases and has supported supervised approaches to text detoxification (Logacheva et al., 2022). Large pretrained language models have also been adapted for detoxification through supervised generation (Dale et al., 2021). At the same time, toxicity classifiers and detoxification systems can exhibit biases or suppress language associated with particular communities, motivating caution when classifier-based objectives are used as automatic measures (Welbl et al., 2021; Xu et al., 2021).

Reinforcement learning and reward optimization. Reinforcement learning makes it possible to optimize generated sequences using rewards that need not be differentiable through the generated text. GRPO was introduced as a memory-efficient alternative to value-model-based policy optimization and estimates advantages by comparing rewards among multiple completions sampled for the same prompt (Shao et al., 2024). The present study does not propose a new RL algorithm. Instead, GRPO is used as an optimization framework for examining how different reward formulations affect the balance between detoxification and semantic preservation. Sentence-level semantic similarity is measured with Sentence-BERT-style embeddings (Reimers and Gurevych, 2019), providing a complementary signal to the toxicity reward.

3 Method

Figure 1 summarizes the experimental pipeline. The SFT model is trained on ParaDetox source-reference pairs, after which two independent GRPO branches are initialized from the same supervised checkpoint. The branches differ only in their reward formulation and are evaluated using the same held-out source-level test set.

3.1 Task and Dataset

ParaDetox (Logacheva et al., 2022) contains 19,744 toxic-neutral source-reference pairs. Importantly, some toxic source sentences are associated with

multiple neutral references. A random split at the individual pair level would therefore allow the same toxic source to appear in more than one split, creating source-level leakage.

To prevent this, the dataset is split at the *toxic-source level*. All neutral references corresponding to the same toxic source are assigned to the same partition. Using a fixed random seed of 42 and an 80/10/10 split over unique toxic sources produces 9,541 training sources, 1,192 validation sources, and 1,194 test sources. Because sources may have multiple references, these partitions correspond to 15,765, 1,932, and 2,047 source-reference pairs, respectively.

The distinction between sources and pairs is important for the experimental setup. Supervised fine-tuning retains all 15,765 training source-reference pairs, treating different human neutralizations of the same source as separate supervised examples. RL training instead operates on the 9,541 unique training sources because the reward is computed directly from model-generated outputs and does not require a reference target.

Final evaluation is similarly source-based. Each of the 1,194 unique held-out toxic sources is evaluated exactly once, so sources with multiple ParaDetox references do not receive additional weight. Human neutral references are therefore used for supervised training but not for the final automatic metrics.

3.2 Supervised Fine-Tuning

The base model is Qwen2.5-0.5B-Instruct (Yang et al., 2024). Parameter-efficient fine-tuning is performed using Low-Rank Adaptation (LoRA) (Hu et al., 2022). Each toxic source is formatted as an instruction to rewrite the sentence into a neutral and non-toxic form while preserving its meaning, and the corresponding ParaDetox neutral reference serves as the target response.

The system instruction is: You are a text rewriting assistant. Your task is to remove toxic, rude, or offensive language while preserving the original meaning as much as possible.

The user instruction is: Rewrite the following sentence in a neutral and non-toxic style. Preserve the original meaning and output only the rewritten sentence.

Training uses assistant-only language-model loss so that only target response tokens contribute to the supervised objective. LoRA is applied to the query, key, value, and output projection layers with

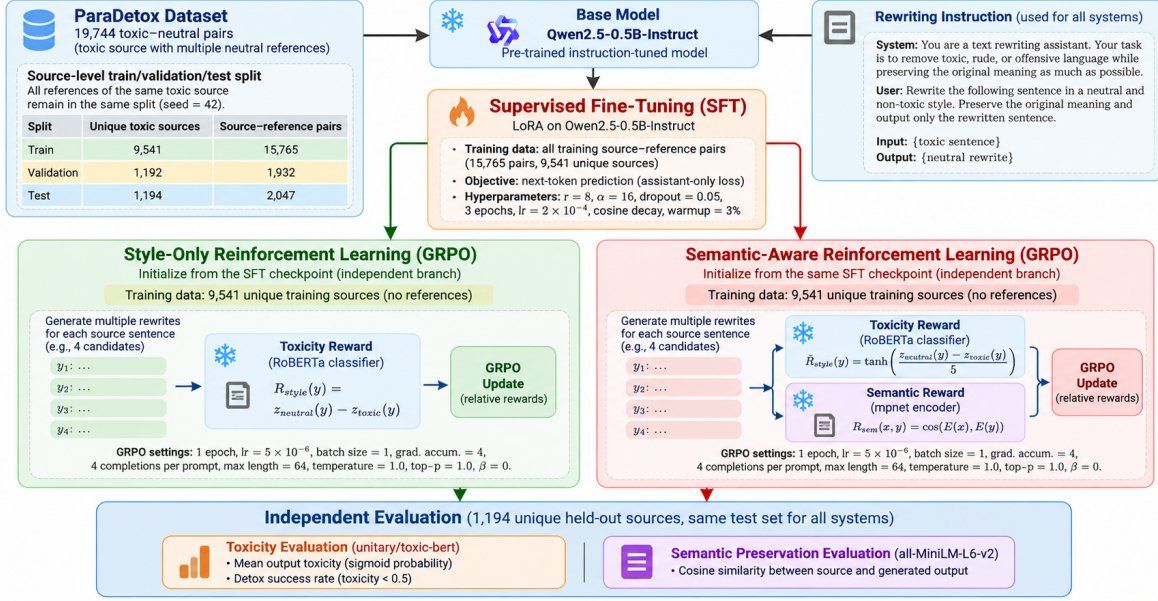


Figure 1: Overview of the experimental pipeline. ParaDetox is split at the toxic-source level to prevent the same source from appearing across training, validation, and test partitions. SFT uses all training source-reference pairs, whereas both GRPO conditions operate on unique toxic training sources and are initialized independently from the same SFT checkpoint. Style-only GRPO optimizes a toxicity-classifier reward, while Style+Semantic GRPO combines a normalized toxicity reward with source-output semantic similarity. Final evaluation uses separate frozen models for toxicity and semantic preservation on the same 1,194 unique held-out sources.

rank $r = 8$, $\alpha = 16$, and dropout 0.05. The model is trained for three epochs with a learning rate of 2×10^{-4} , cosine decay, a warmup ratio of 0.03, maximum sequence length 256, per-device batch size 1, and gradient accumulation over four steps. The checkpoint with the lowest validation loss is retained; the best validation loss is 0.8651.

3.3 Style-Only Reinforcement Learning

The first RL condition is initialized directly from the best SFT checkpoint. GRPO samples four candidate completions for each toxic source and optimizes their relative rewards.

A frozen RoBERTa toxicity classifier, `s-nlp/roberta_toxicity_classifier`, provides the style reward. Let $z_{neutral}$ and z_{toxic} denote its neutral and toxic logits for a generated output $\hat{y}$. The style reward is

$$R_{style}(\hat{y}) = z_{neutral}(\hat{y}) - z_{toxic}(\hat{y}). \quad (1)$$

This logit-margin objective directly favors outputs classified as neutral. No semantic-preservation component is included in this condition, making it possible to examine the behavior produced by optimizing detoxification alone.

Training runs for one epoch over the 9,541 unique training sources with a learning rate of

5×10^{-6} , cosine scheduling, batch size 1, gradient accumulation of 4, four sampled generations per prompt, maximum completion length 64, temperature 1.0, and top- $p = 1.0$. The GRPO KL coefficient is set to $\beta = 0$.

3.4 Semantic-Aware Reinforcement Learning

The second RL condition is initialized independently from the same best SFT checkpoint rather than from the style-only RL model. This parallel design isolates the effect of the reward function from sequential RL training.

Semantic preservation is measured during RL training using `all-mpnet-base-v2`. For toxic source x and generated output $\hat{y}$, the semantic reward is

$$R_{sem}(x, \hat{y}) = \cos(E_{mpnet}(x), E_{mpnet}(\hat{y})). \quad (2)$$

Initial experiments revealed a substantial scale mismatch between the raw style logit margin and cosine similarity. For example, an early diagnostic batch produced a style-reward mean of 6.802 with standard deviation 2.679, compared with a semantic-reward mean of 0.781 with standard deviation 0.164. A nominally equal weighted sum would therefore still be dominated by the style reward.

To reduce this imbalance, only the style component is bounded:

$$\tilde{R}_{\text{style}}(\hat{y}) = \tanh\left(\frac{z_{\text{neutral}}(\hat{y}) - z_{\text{toxic}}(\hat{y})}{5}\right). \quad (3)$$

The final combined reward is

$$R_{\text{combined}}(x, \hat{y}) = 0.5 \tilde{R}_{\text{style}}(\hat{y}) + 0.5 R_{\text{sem}}(x, \hat{y}). \quad (4)$$

After normalization, a diagnostic batch produced means of 0.800 for the style component and 0.777 for the semantic component, giving the two terms substantially more comparable numerical scales. All other GRPO settings are kept consistent with the style-only condition.

4 Experimental Setup

Four systems are evaluated: zero-shot prompting of the base Qwen2.5-0.5B-Instruct model, SFT-LoRA, style-only GRPO, and Style+Semantic GRPO. All systems use the same rewriting instruction. Final generation is deterministic with greedy decoding (do_sample=False) and a maximum of 64 new tokens. Evaluation is performed on the 1,194 unique toxic sources in the held-out test set.

Semantic preservation. The final semantic evaluator is sentence-transformers/all-MiniLM-L6-v2. Importantly, semantic similarity is computed directly between the original toxic source x and the generated rewrite $\hat{y}$:

$$S_{\text{sem}}(x, \hat{y}) = \cos(E(x), E(\hat{y})). \quad (5)$$

The ParaDetox neutral references are not used for this metric. Source–output comparison measures how much semantic content from the original sentence is retained without requiring the generated output to reproduce any particular human neutralization.

Toxicity. A separate frozen classifier, unitary/toxic-bert, is used for final toxicity evaluation. This evaluator is distinct from the RoBERTa model used to construct the RL style reward, reducing direct dependence between the optimization reward and the reported toxicity metric. Mean output toxicity is reported, together with *detox success*, defined as

$$\text{DetoxSuccess} = \frac{1}{N} \sum_{i=1}^N I[T(\hat{y}_i) < 0.5]. \quad (6)$$

Statistical analysis. Because every system generates one output for the same 1,194 test sources, statistical comparisons are paired by source. Test-set uncertainty is estimated using 10,000 paired bootstrap resamples with seed 42. Percentile 95% confidence intervals are computed for aggregate metrics and for pairwise differences between systems. For binary detoxification success, exact McNemar tests additionally compare paired success/failure outcomes.

These analyses quantify uncertainty associated with the fixed test examples. They do not measure variability across independently trained models, since each training condition is run only once.

5 Results and Discussion

5.1 Quantitative Results

Table 1 summarizes the main automatic evaluation. Zero-shot prompting provides moderate semantic preservation but leaves substantial residual toxicity. SFT produces a large improvement in detoxification while also increasing semantic similarity.

Style-only GRPO strongly optimizes the toxicity objective. Mean output toxicity falls from 0.0685 under SFT to 0.0033, while detoxification success increases from 95.14% to 100.00%. However, this improvement is accompanied by a severe decline in semantic similarity, from 0.7983 to 0.4153. The paired bootstrap difference in semantic similarity between SFT and style-only GRPO is 0.3830 with a 95% confidence interval of [0.3709, 0.3951]. The corresponding style-only improvement in detox success over SFT is 4.86 percentage points (95% CI: [3.69, 6.11] percentage points).

The combined reward substantially changes this behavior. Style+Semantic GRPO reaches a semantic similarity of 0.8657, compared with 0.7983 for SFT and 0.4153 for style-only GRPO. Relative to SFT, the observed improvement in semantic similarity is 0.0674, with a paired bootstrap 95% confidence interval of [0.0603, 0.0748]. Relative to style-only GRPO, the improvement is 0.4505 (95% CI: [0.4391, 0.4620]).

The Style+Semantic model achieves an observed detoxification success rate of 94.30%, compared with 95.14% for SFT. The SFT-minus-Style+Semantic difference is 0.84 percentage

Method	Semantic $\uparrow$	Toxicity $\downarrow$	Detox $\uparrow$
Zero-shot	.7268	.3979	58.88%
SFT-LoRA	.7983	.0685	95.14%
Style-only GRPO	.4153	.0033	100.00%
Style+Semantic GRPO	.8657	.0774	94.30%

Table 1: Automatic evaluation on 1,194 unique ParaDetox test sources. Semantic similarity is computed between each toxic source and its generated rewrite. Detox success is the percentage of generated outputs with toxicity below 0.5. Statistical uncertainty is analyzed using paired bootstrap resampling and exact McNemar tests.

points, with a paired bootstrap 95% confidence interval of $[-0.25, 1.93]$ percentage points. An exact McNemar test ($p = .164$) likewise provides no evidence of a difference in binary detoxification success under this fixed test-set comparison.

Mean toxicity nevertheless reveals a small difference not captured by the binary threshold alone. SFT obtains 0.0685 compared with 0.0774 for Style+Semantic GRPO. The paired SFT-minus-Style+Semantic difference is -0.0089 , with a 95% bootstrap confidence interval of $[-0.0159, -0.0020]$. Thus, the combined RL objective produces slightly higher average toxicity scores even though the binary detox success rates remain close. Additional results see Appendix A.

Overall, the quantitative results answer the research question in a reward-dependent manner. Optimizing toxicity alone can improve detoxification beyond SFT, but in this experiment it does so at a substantial semantic cost. Adding an explicit semantic reward prevents this collapse and produces the highest source–output semantic similarity while retaining detoxification performance close to the supervised baseline. The experiment therefore does not support a general claim that RL uniformly outperforms SFT; rather, it shows that the behavior produced by RL depends strongly on the structure and scale of the reward.

5.2 Qualitative Analysis

Qualitative inspection helps explain the semantic degradation under style-only GRPO. For example, the source “*as far as the q situation they ain’t got they shit together*” is rewritten as “*As far as the situation, it seems that things have improved for them.*”, reversing the original negative assessment. Style+Semantic GRPO instead produces “*as far as the q situation they ain’t got they together*”, which is less fluent but better preserves the original

proposition.

A similar semantic change occurs for “*sumbody hacked my damn twitter*”. Style-only GRPO generates “*Someone has been monitoring my Twitter posts for me.*”, changing the reported event, whereas Style+Semantic GRPO produces “*sumbody hacked my twitter*”. These failures are consistent with reward over-optimization: optimizing only the toxicity reward does not directly protect semantic faithfulness.

The combined reward is not perfect either. For example, “*lol stay ya ass in the car*” becomes “*Lol stay ya in the car*”, showing that toxicity removal can still reduce grammaticality. Thus, higher embedding similarity should not be interpreted as guaranteeing semantic correctness for every rewrite.

6 Conclusion and Limitations

This study investigated whether reinforcement learning can improve text detoxification beyond supervised fine-tuning without sacrificing semantic preservation. Style-only GRPO achieves stronger detoxification but causes substantial semantic degradation, with qualitative examples showing factual changes and content loss. Adding a semantic reward largely prevents this failure mode, producing the highest semantic similarity while maintaining detoxification performance close to the SFT baseline. Overall, the results show that the effectiveness of RL for controllable generation depends strongly on reward design: optimizing toxicity alone can encourage behavior consistent with reward over-optimization, while explicitly rewarding semantic preservation leads to a better balance between the two objectives.

Several limitations remain. Each training condition is run only once, so the paired statistical analysis captures test-set uncertainty rather than variation across independent training runs. In addition, embedding-based semantic similarity may miss factual or subtle meaning changes, while classifier-based toxicity evaluation cannot fully represent human judgments. The experiments use $\beta = 0$ for GRPO; evaluating whether a non-zero KL penalty can further reduce semantic drift then is left for future work. Finally, the experiments are limited to one dataset and one relatively small base model. Future work should include multiple training seeds, larger models, alternative reward formulations, and human evaluation.

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Code and models are available at: <https://github.com/Yuzhou0210/style-rl.git>

A Additional Statistical Results

Method	Semantic Similarity $\uparrow$	Output Toxicity $\downarrow$	Detox Success $\uparrow$
Zero-shot	.7268 [.7140, .7399]	.3979 [.3741, .4233]	58.88 [56.03, 61.64]%
SFT-LoRA	.7983 [.7888, .8078]	.0685 [.0590, .0787]	95.14 [93.89, 96.31]%
Style-only GRPO	.4153 [.4040, .4265]	.0033 [.0023, .0045]	100.00 [100.00, 100.00]%
Style+Semantic GRPO	.8657 [.8590, .8723]	.0774 [.0669, .0882]	94.30 [92.96, 95.56]%

Table 2: Aggregate metrics with 95% paired-bootstrap confidence intervals over 10,000 resamples of the 1,194 test sources. The degenerate bootstrap interval for Style-only GRPO detox success results from all observed test outputs satisfying the toxicity threshold.

Comparison	Metric	Difference	95% CI
SFT – Style-only	Semantic	+.3830	[.3709, .3951]
	Toxicity	+.0652	[.0557, .0753]
	Detox	−4.86 pp	[−6.11, −3.69]
SFT – Style+Semantic	Semantic	−.0674	[−.0748, −.0603]
	Toxicity	−.0089	[−.0159, −.0020]
	Detox	+0.84 pp	[−0.25, +1.93]
Style-only – Style+Semantic	Semantic	−.4505	[−.4620, −.4391]
	Toxicity	−.0741	[−.0849, −.0637]
	Detox	+5.70 pp	[+4.44, +7.04]

Table 3: Paired differences and 95% bootstrap confidence intervals. Differences are defined as the first method minus the second method.

McNemar Test. For SFT and Style+Semantic GRPO, 42 test examples have discordant binary detox outcomes: SFT succeeds while Style+Semantic fails on 26 examples, whereas Style+Semantic succeeds while SFT fails on 16. An exact McNemar test gives $p = .164$. In comparison, Style-only GRPO achieves successful detoxification on all 1,194 test examples.

B AI Usage Statement

Generative AI tools were used as assistive tools during the development and writing of this work. Specifically, they were used to help revise and debug code, identify potential implementation issues, and improve the clarity, grammar, and presentation of the manuscript. All research questions, task formulations, experimental designs, model and training configurations, evaluation protocols, and interpretation of experimental results were determined by the author. The core implementation and experimental pipeline were developed and validated by the author, who also reviewed and verified all AI-assisted code modifications and written content. Generative AI was not used to generate experimental results or to make research decisions autonomously.